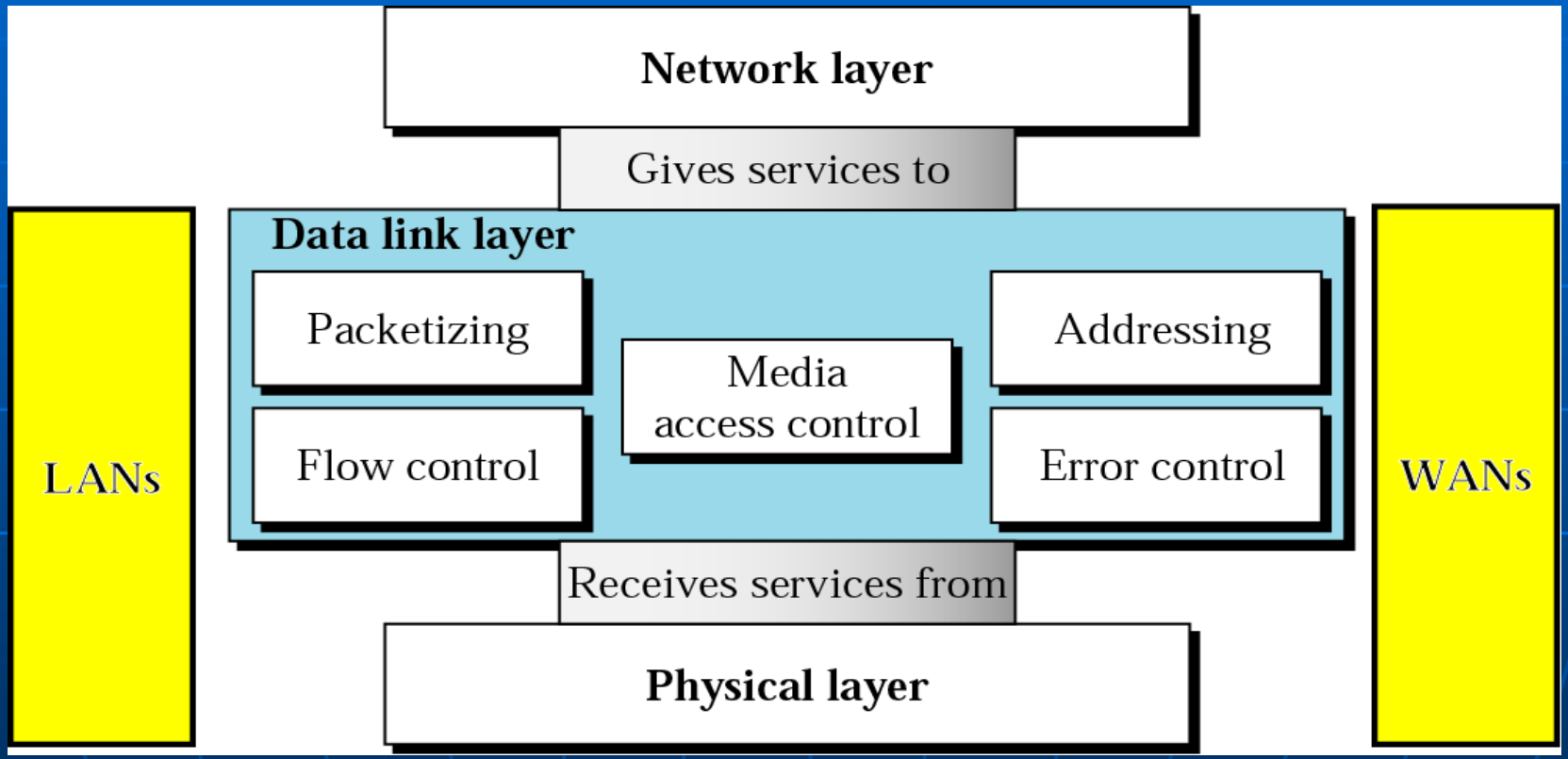
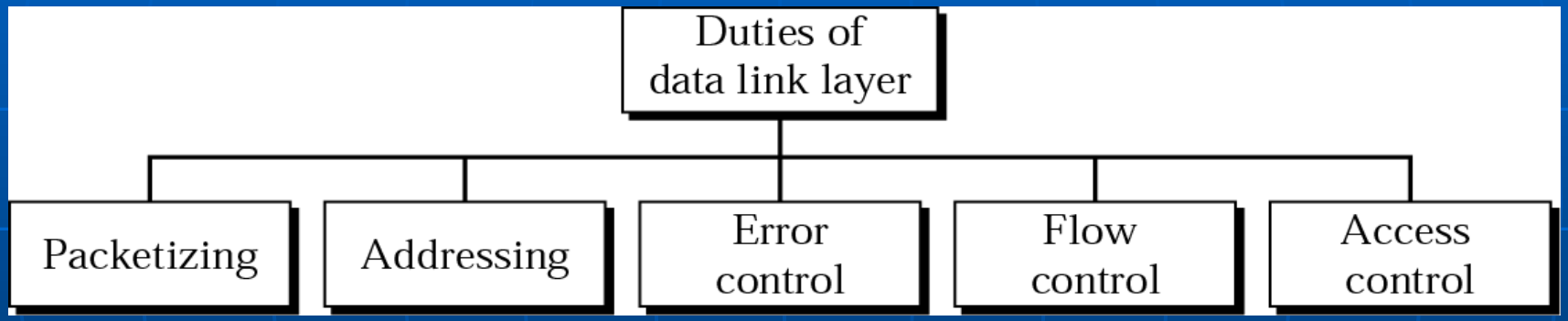
Data Link Layer

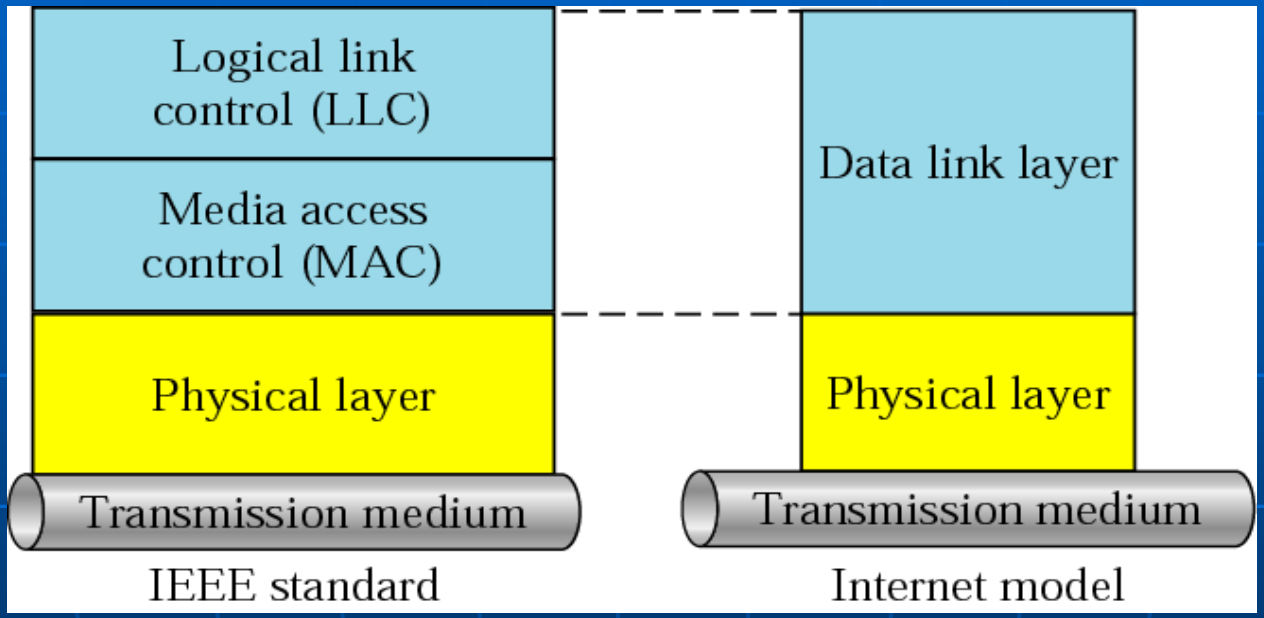
Position of the data-link layer



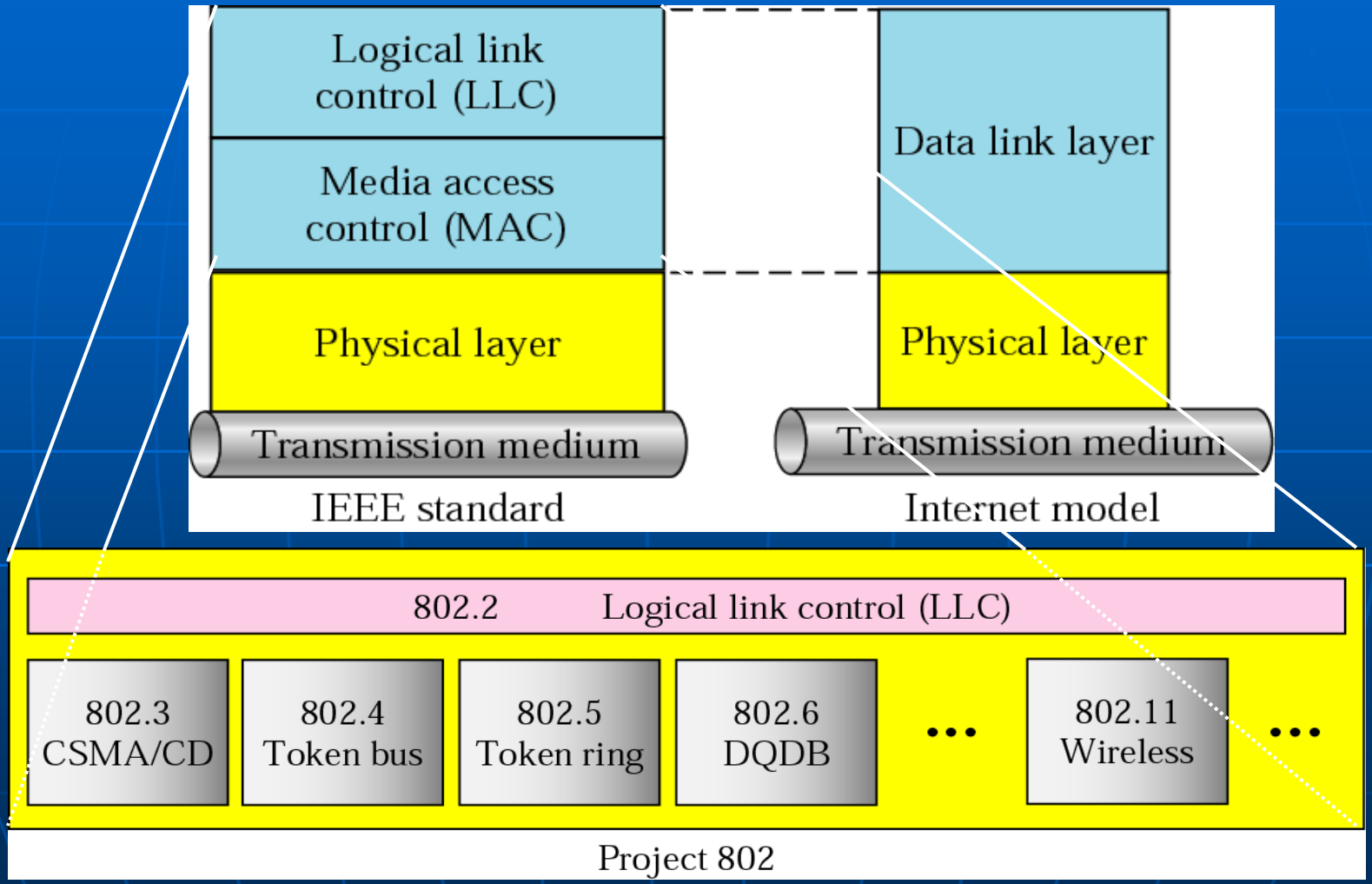
Data link layer duties



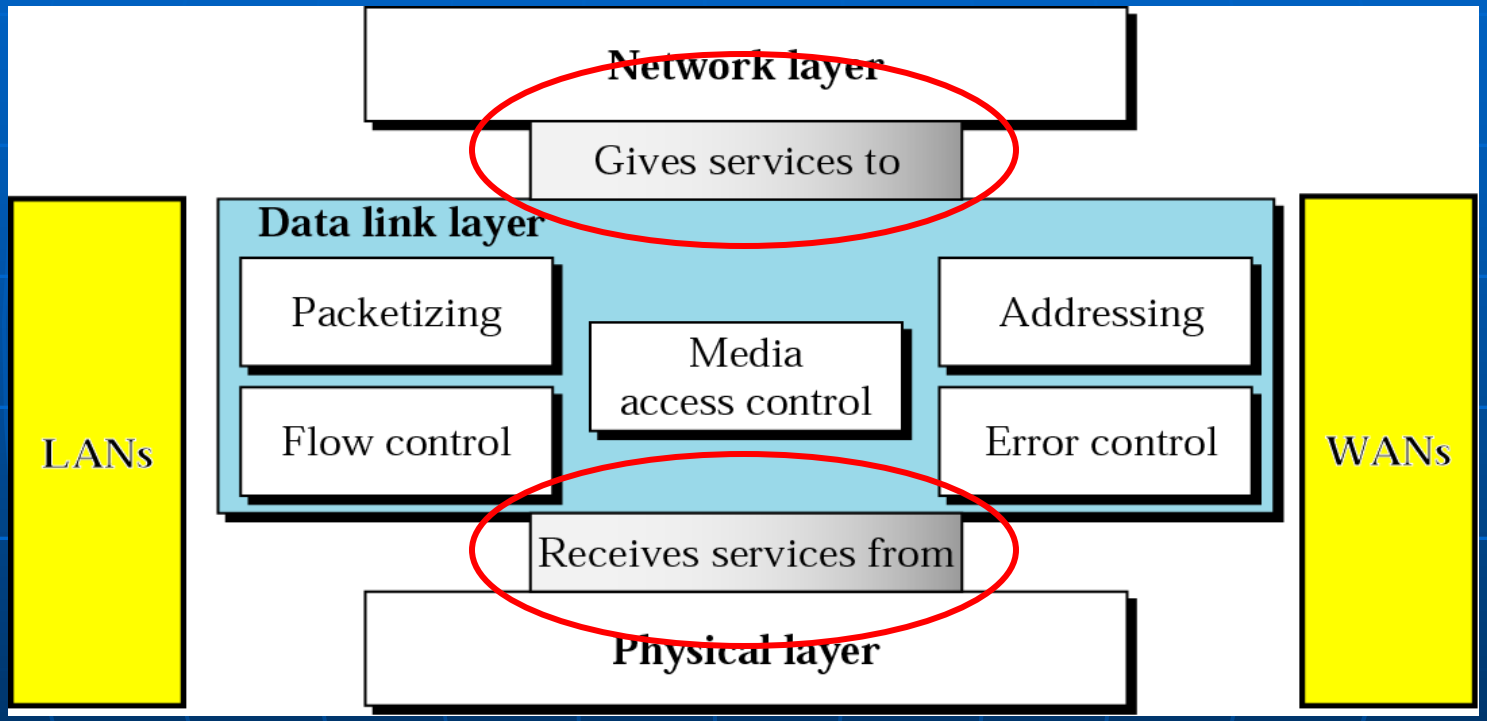
LLC and MAC sublayers



IEEE standard



Data Link layer



Data Link Layer

- The main tasks of the data link layer are:
  - Transfer data from the network layer of one machine to the network layer of another machine
  - Convert the raw bit stream of the physical layer into groups of bits ("frames")



Services Provided by the Data link Layer

- receives services from Physical layer and provides service to the network layer
- Reliable transfer of DL-SDU (Data link service data unit) to the network layer over the data link connection which is established, maintained and release
- Sequencing
- Error Notification
- Flow control
- QoS parameter-Residual error rate,transit delay,throughput,

Data Link Protocols

- Binary Synchronous Data Link Control (BISYNC,BSC)
- Synchronous Data link Control (SDLC)
- High Level Data Link Control (HDLC)
- Advanced data Communication Control Procedure (ADCCP)

Flow Control Policies

- Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment

- Feed-back based
  - communicate receiver state to sender
    - so sender can take decision
  - explicit start/stop commands
  - ask for another fixed maximum amount of data
    - more smoothly
- Rate-based
  - internal mechanism to adjust the rate

- Data-Link: mainly feed-back based methods

11.1 Flow and Error Control

Flow Control

Error Control



Flow control refers to a set of procedures used to restrict the amount of data that the sender can send before waiting for acknowledgment.



Note:

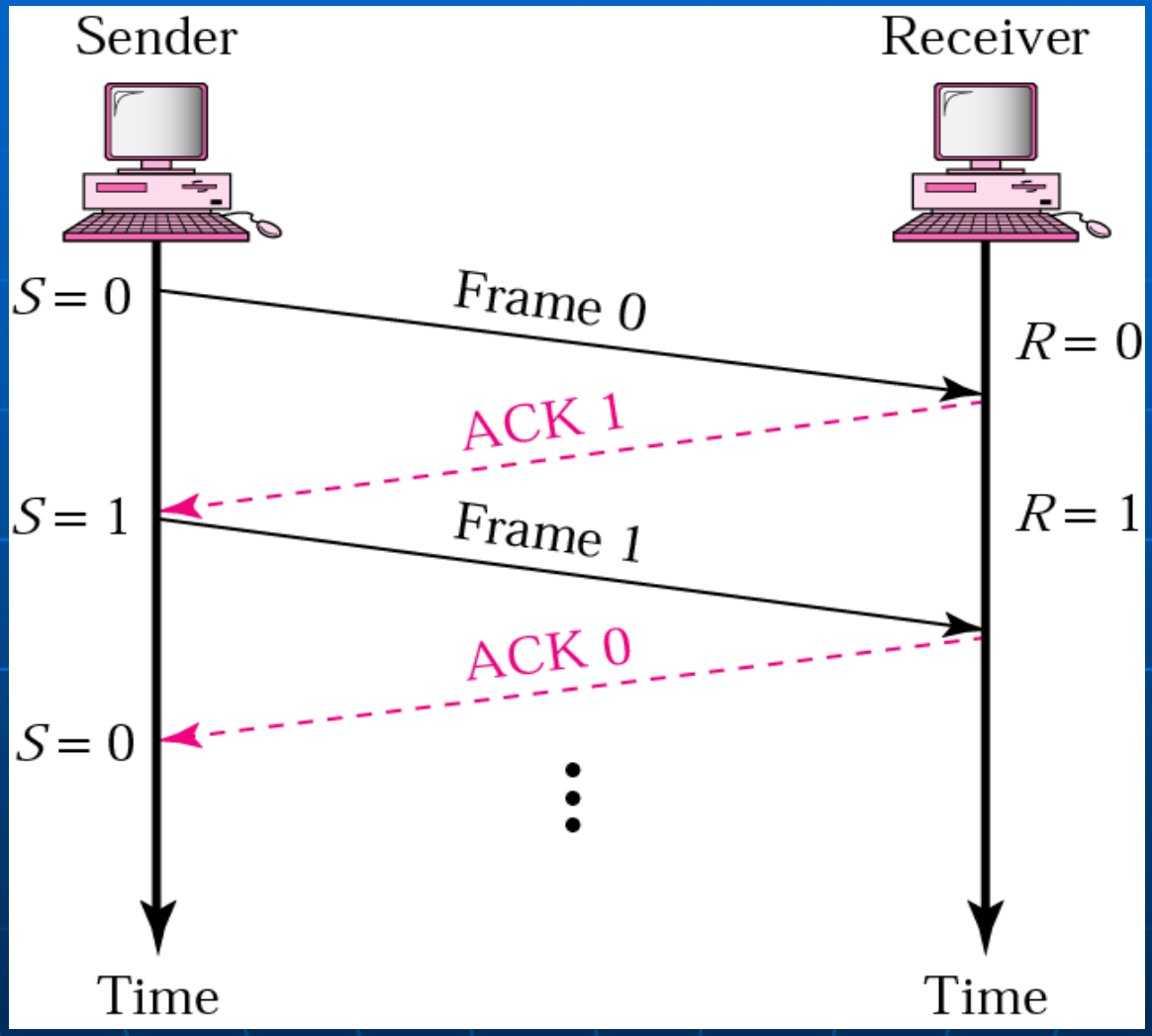
Error control in the data link layer is based on automatic repeat request, which is the retransmission of data.

11.2 Stop-and-Wait ARQ

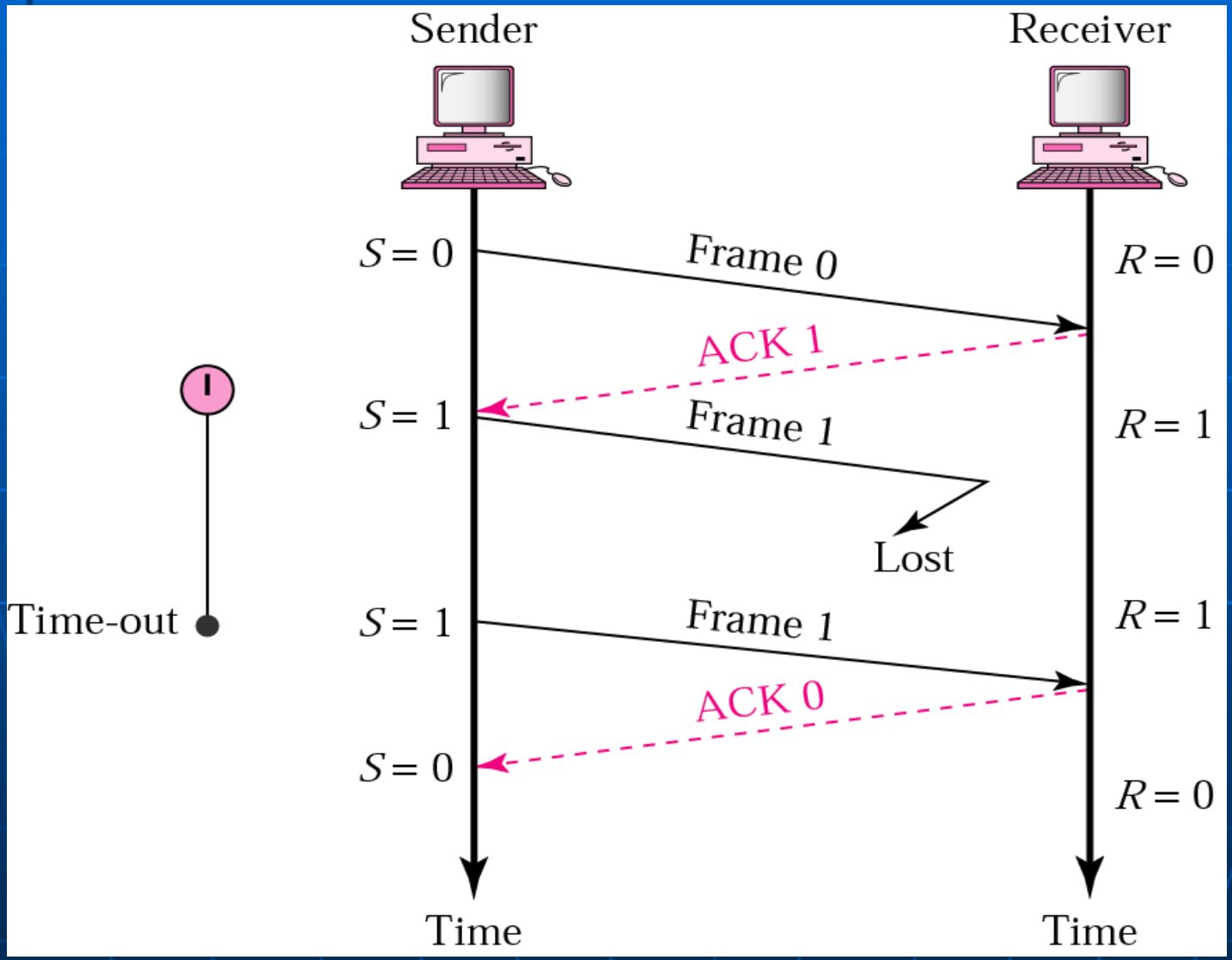
Operation

Bidirectional Transmission

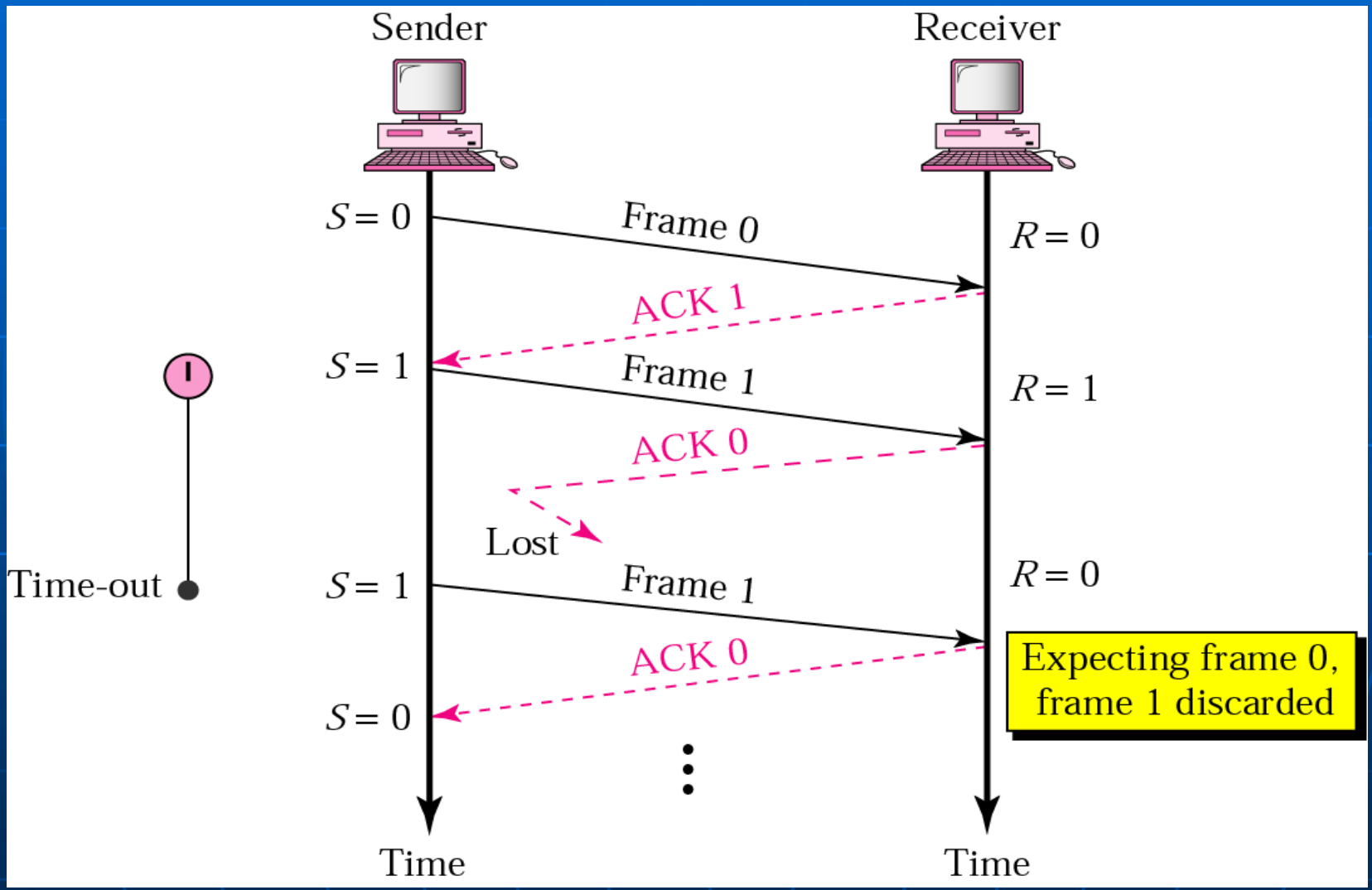
Normal operation



Stop-and-Wait ARQ, lost frame



Stop-and-Wait ARQ, lost ACK frame

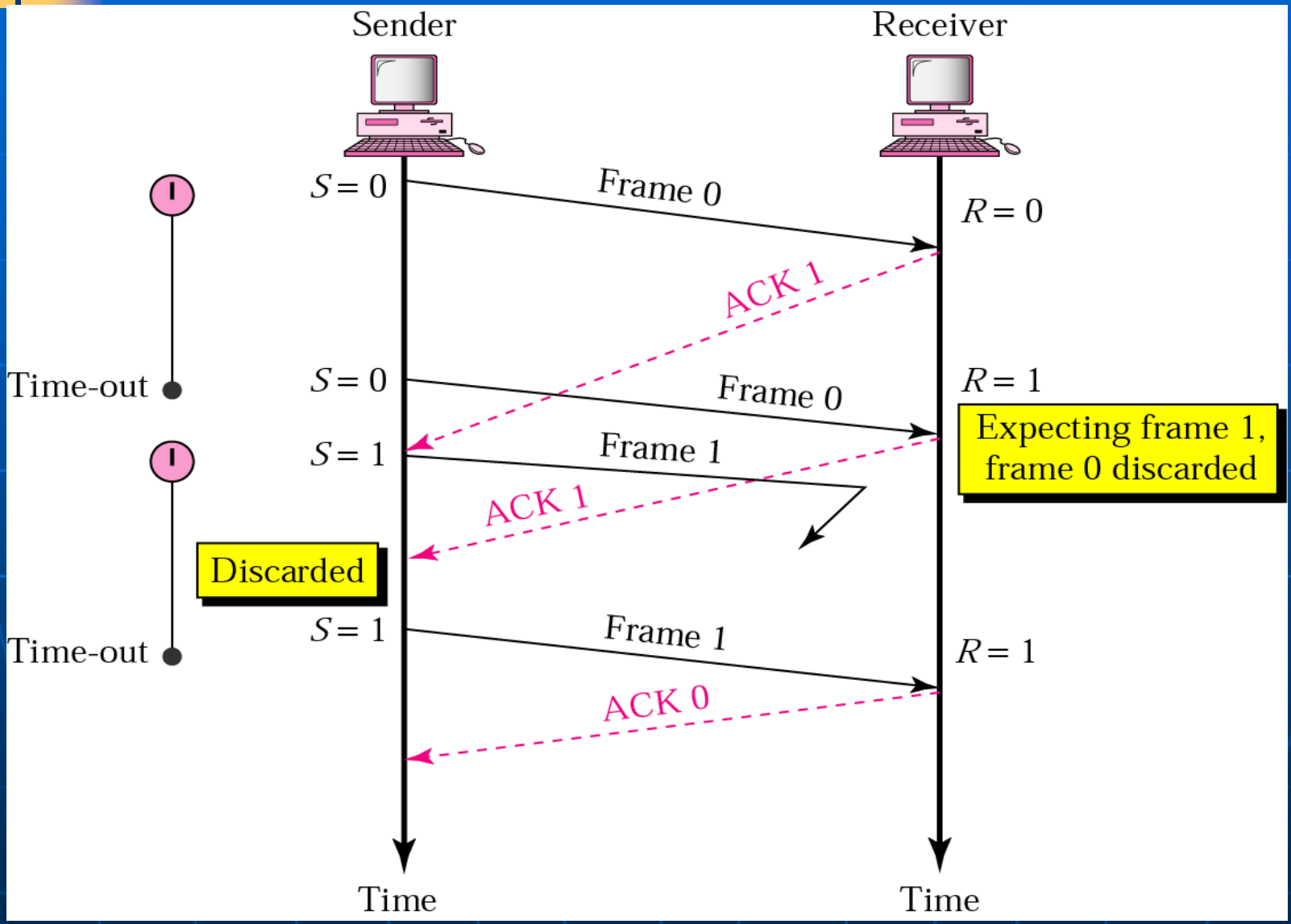




Note:

In Stop-and-Wait ARQ, numbering frames prevents the retaining of duplicate frames.

Stop-and-Wait ARQ, delayed ACK

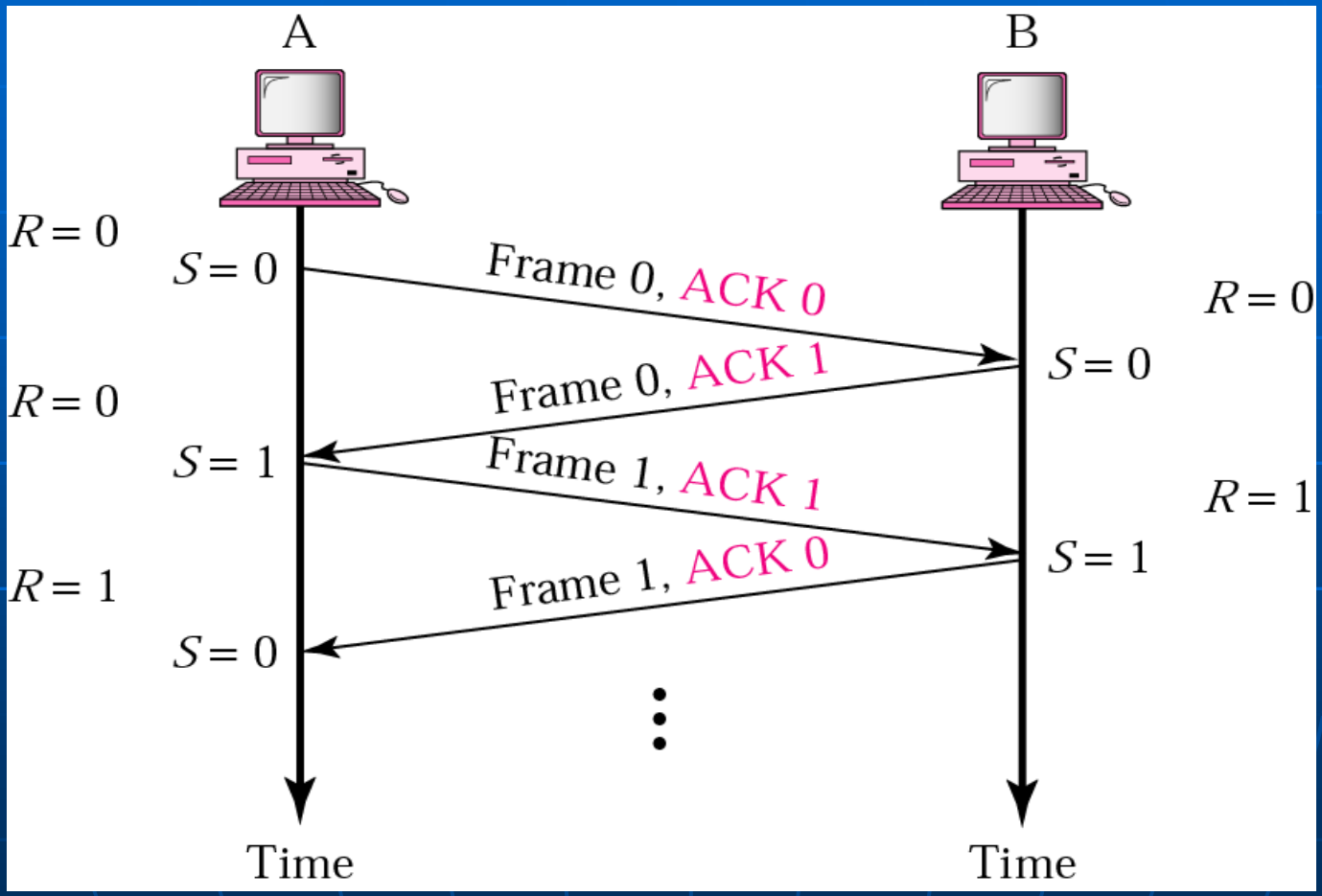




Note:

Numbered acknowledgments are needed if an acknowledgment is delayed and the next frame is lost.

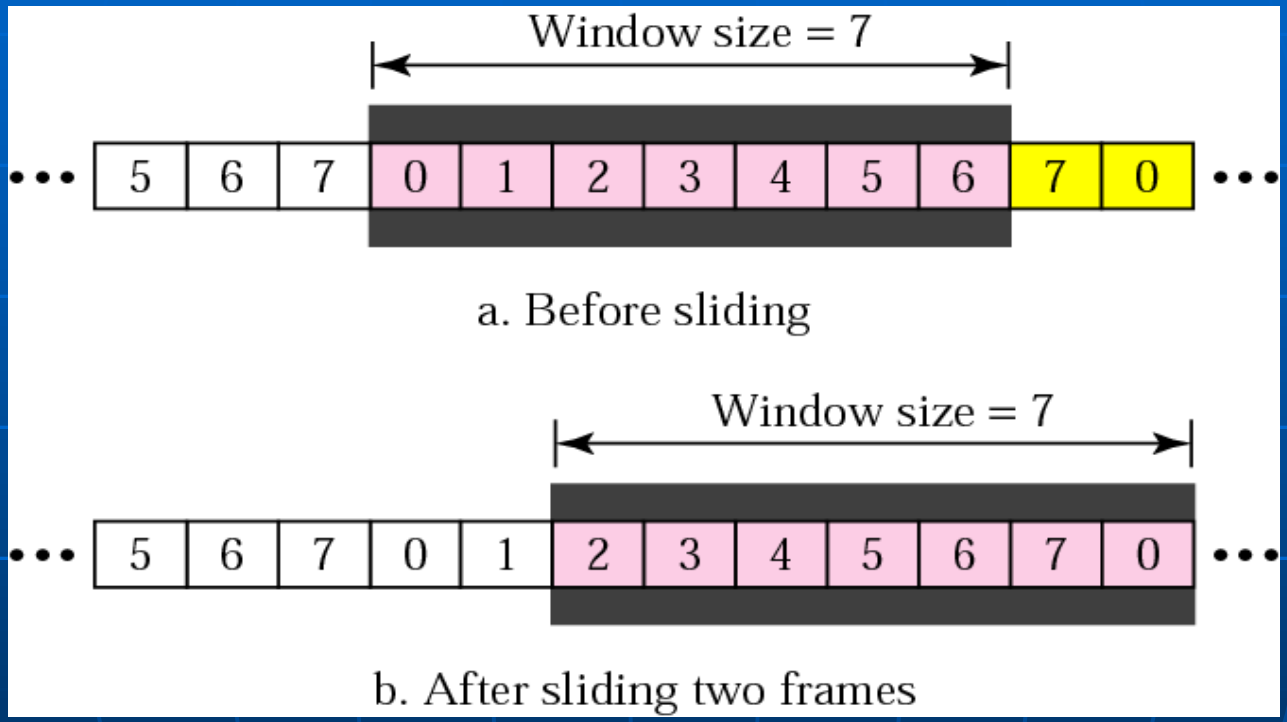
Piggybacking



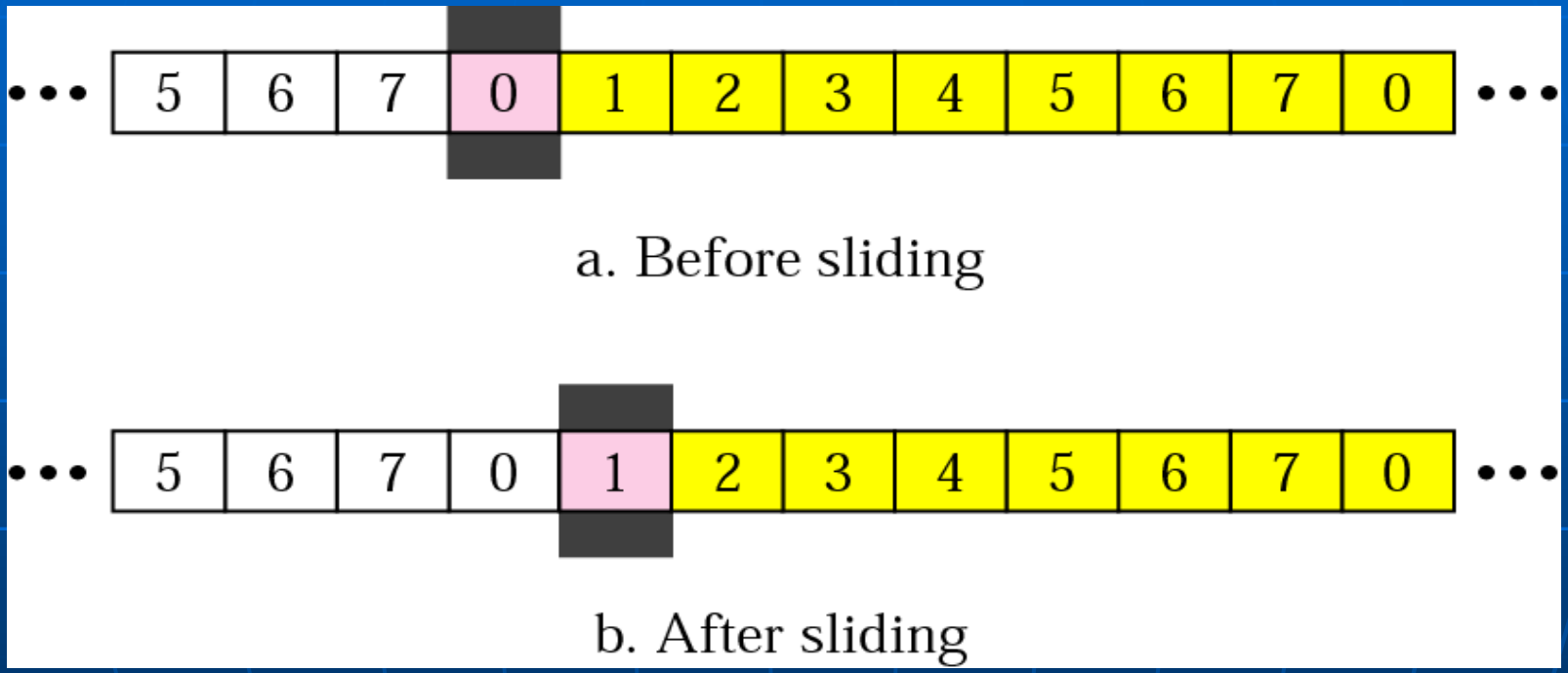
Go-Back-N ARQ

- *Sequence Number*
- *Sender and Receiver Sliding Window*
- *Control Variables and Timers*
- *Acknowledgment*
- *Resending Frames*
- *Operation*

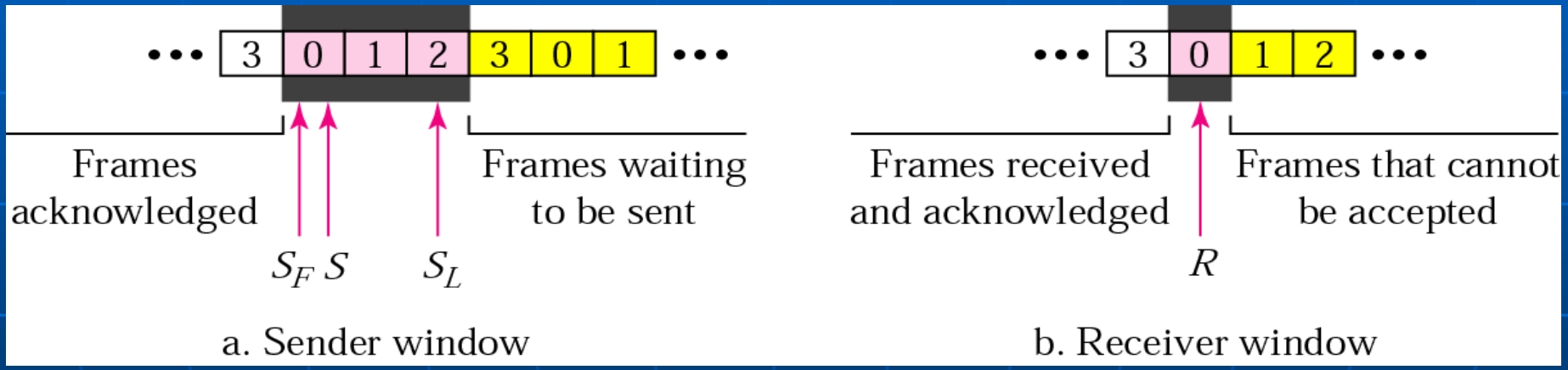
Sender sliding window



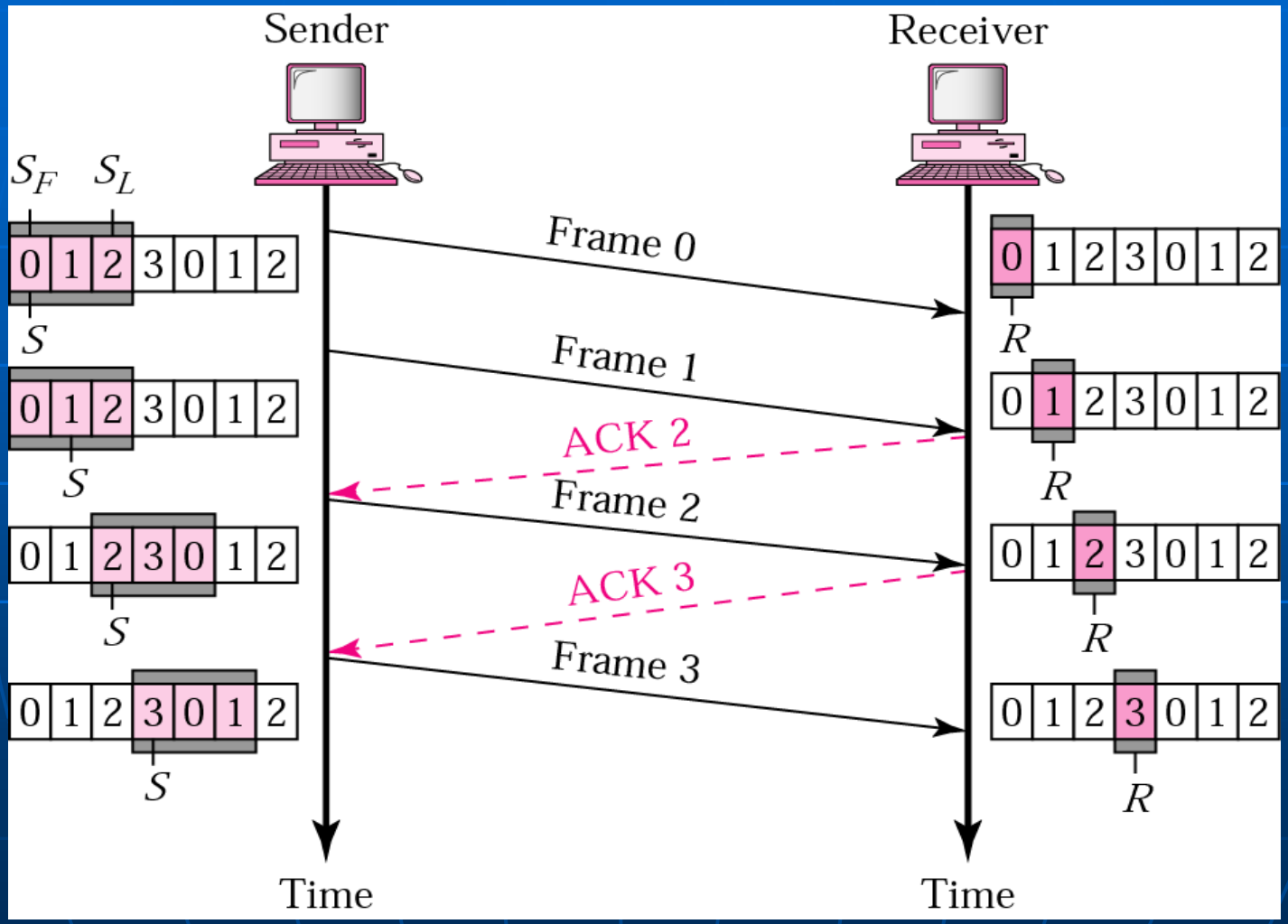
Receiver sliding window



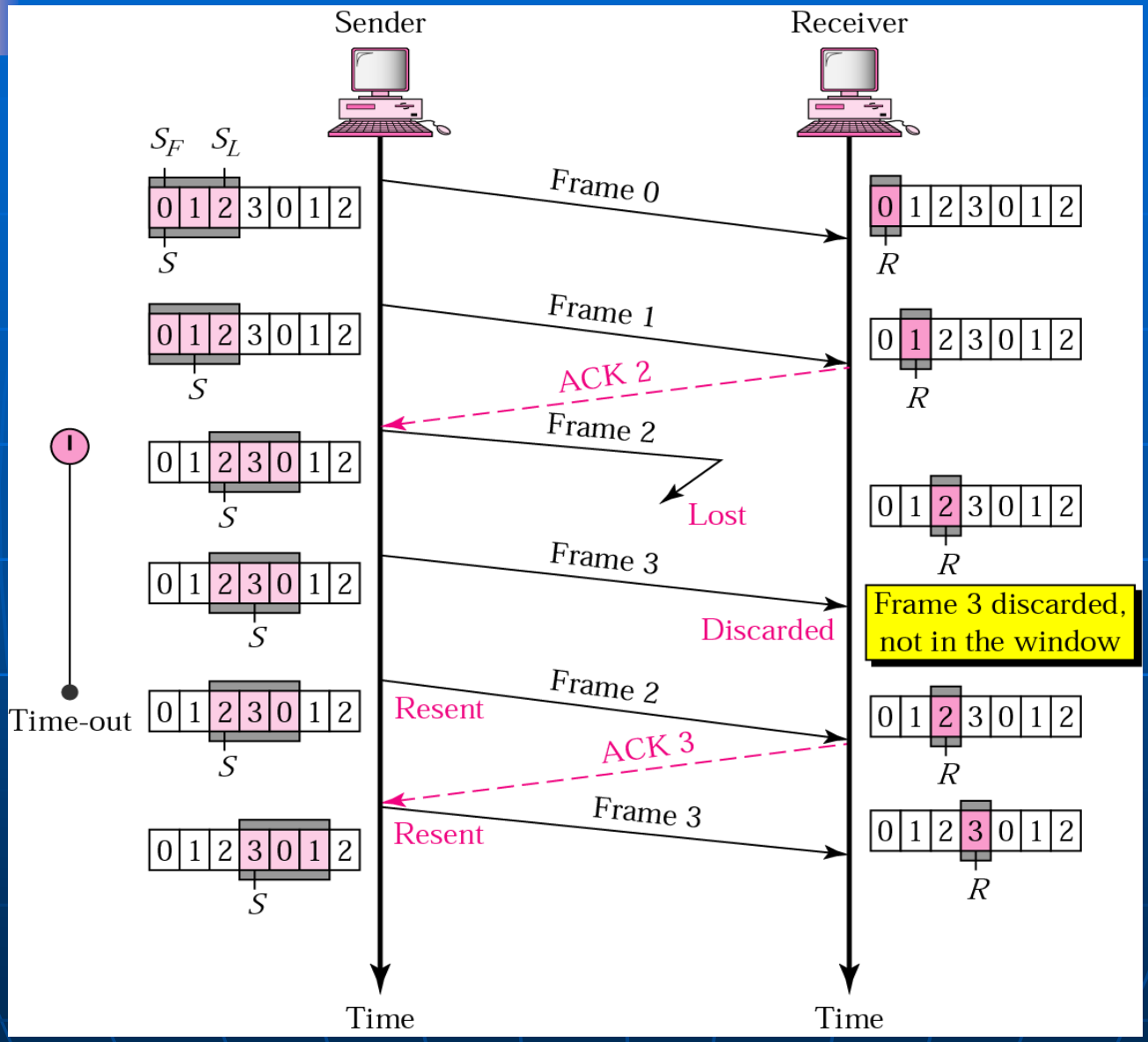
Control variables



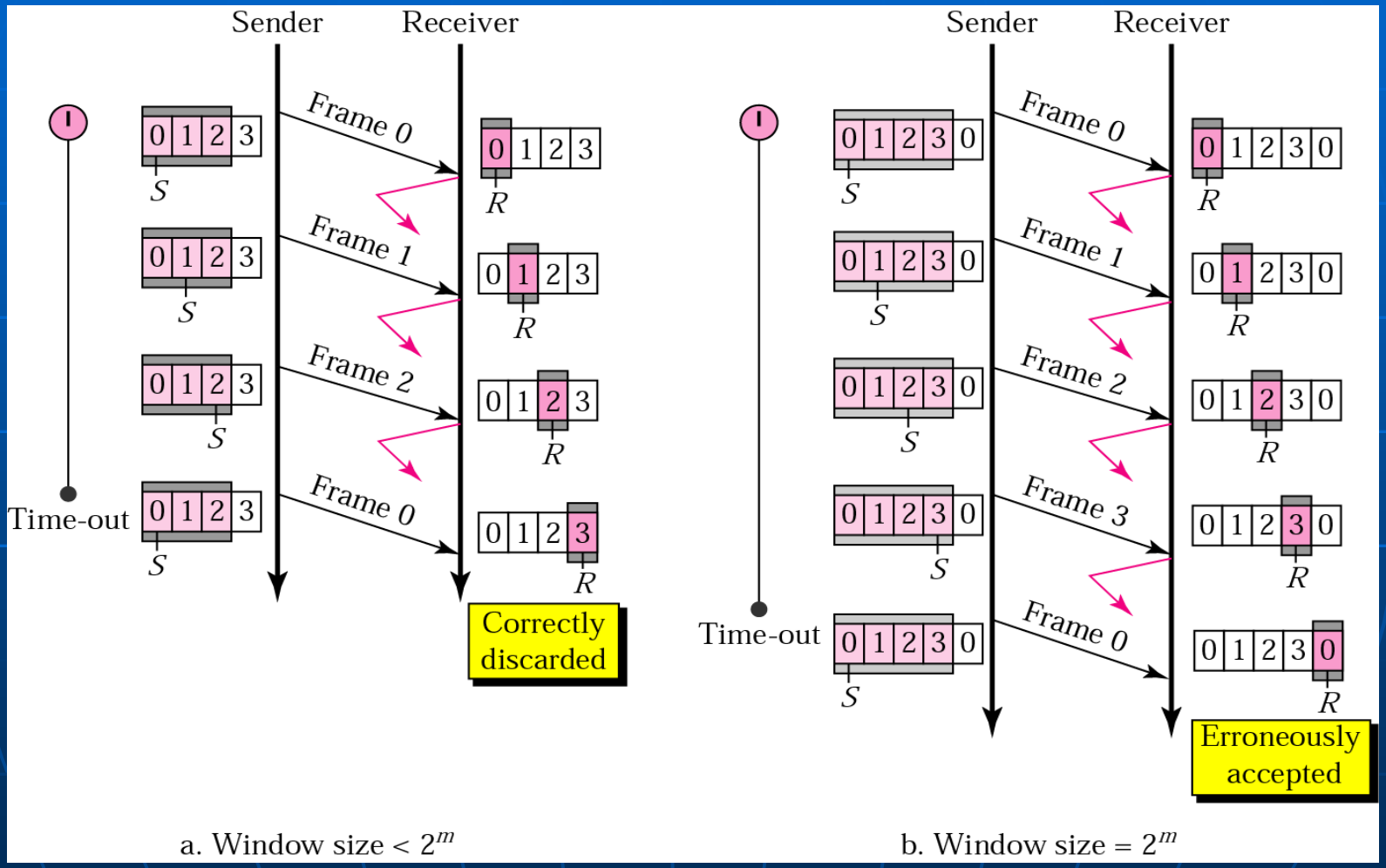
Go-Back-N ARQ, normal operation



Go-Back-N ARQ, lost frame



Go-Back-N ARQ: sender window size





Note:

In Go-Back-N ARQ, the size of the sender window must be less than 2m; the size of the receiver window is always 1.

Selective-Repeat ARQ

Sender and Receiver Windows

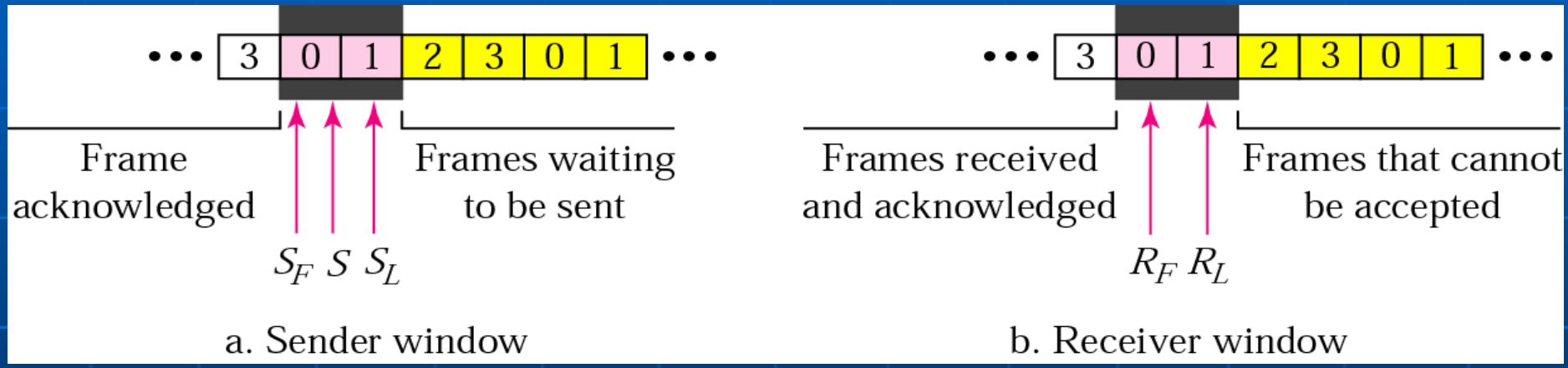
Operation

Sender Window Size

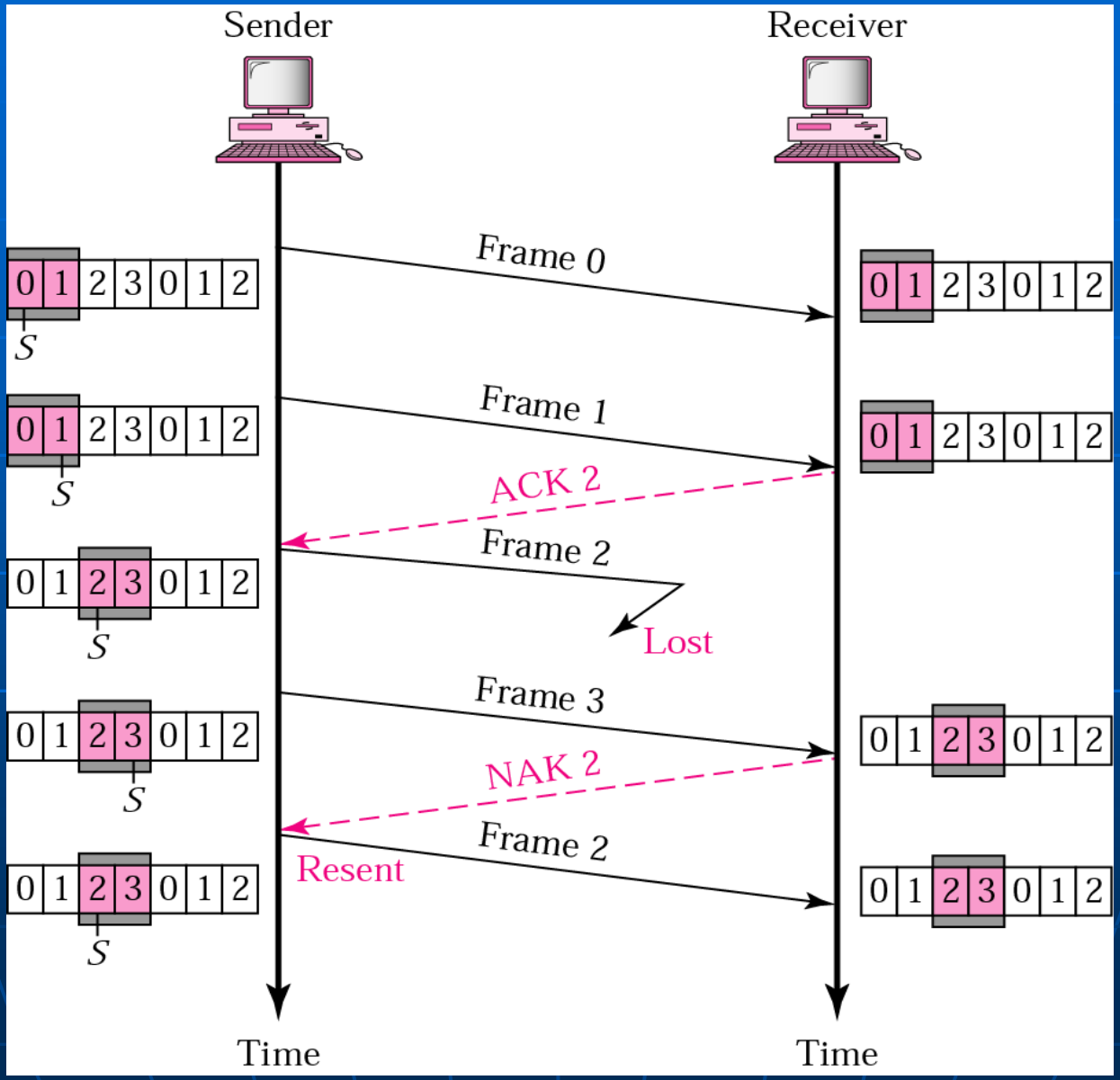
Bidirectional Transmission

Pipelining

11.12 Selective Repeat ARQ, sender and receiver windows



11.13 Selective Repeat ARQ, lost frame

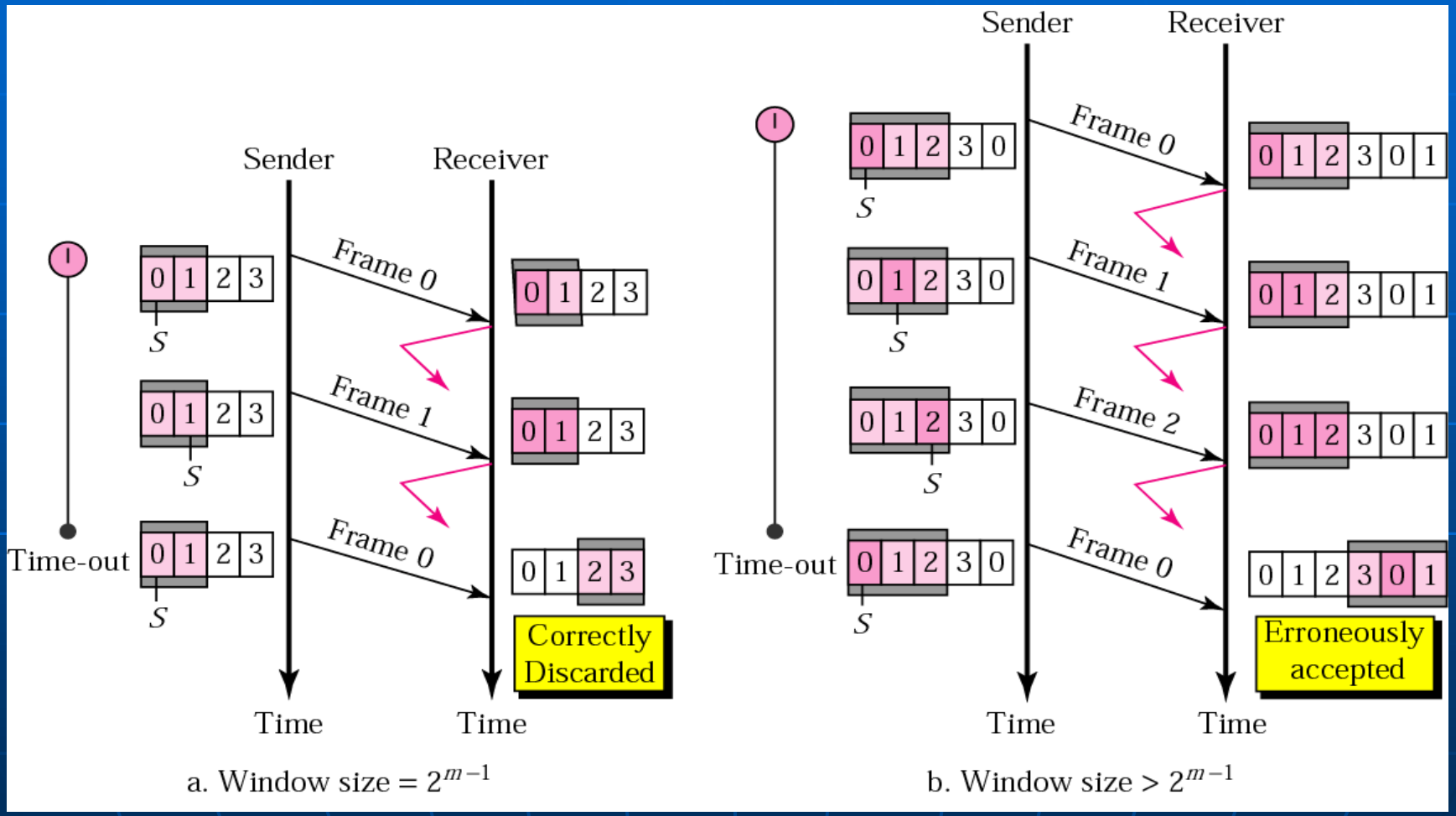




Note:

In Selective Repeat ARQ, the size of the sender and receiver window must be at most one-half of 2^m.

11.14 Selective Repeat ARQ, sender window size



Example 1

In a Stop-and-Wait ARQ system, the bandwidth of the line is 1 Mbps, and 1 bit takes 20 ms to make a round trip. What is the bandwidth-delay product? If the system data frames are 1000 bits in length, what is the utilization percentage of the link?

Solution

The bandwidth-delay product is

$$1 \times 10^6 \times 20 \times 10^{-3} = 20{,}000 \text{ bits}$$

The system can send 20,000 bits during the time it takes for the data to go from the sender to the receiver and then back again. However, the system sends only 1000 bits. We can say that the link utilization is only 1000/20,000, or 5%. For this reason, for a link with high bandwidth or long delay, use of Stop-and-Wait ARQ wastes the capacity of the link.

Example 2

What is the utilization percentage of the link in Example 1 if the link uses Go-Back-N ARQ with a 15-frame sequence?

Solution

The bandwidth-delay product is still 20,000. The system can send up to 15 frames or 15,000 bits during a round trip. This means the utilization is 15,000/20,000, or 75 percent. Of course, if there are damaged frames, the utilization percentage is much less because frames have to be resent.